



Cloud Computing

Assignment-Week 8

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 10

Total mark: 10 X 1 = 10

QUESTION 1:

Which command is used to remove a Docker image that is no longer required?

- A) docker rm
- B) docker rmi
- C) docker delete
- D) docker remove

Correct Answer: B

Detailed Solution: docker rmi removes one or more Docker images from the local system. docker rm is used to remove containers, not images.

QUESTION 2:

Which Docker command is primarily used to list all available Docker networks?

- A) docker network ls
- B) docker network show
- C) docker net ps
- D) docker networks

Correct Answer: A

Detailed Solution: docker network ls displays all Docker networks available on the host. It shows information such as network ID, name, driver, and scope.

QUESTION 3:

Which of the following is the most accurate comparison between VMs and containers?

- A) VMs share the host kernel, while containers have separate kernels
- B) Containers generally share the host kernel, while VMs virtualize hardware and run separate guest OS kernels
- C) Both always require separate guest operating systems
- D) Containers require more resources than VMs because they include a complete OS

Correct Answer: B

Detailed Solution: A VM typically includes a complete guest OS and its own kernel, operating on virtualized hardware. Containers share the host kernel and isolate applications at the operating-system level, making them generally lighter than VMs.



QUESTION 4:

What is the primary role of etcd in a Kubernetes cluster?

- A) Running application containers
- B) Storing Kubernetes cluster state and configuration data
- C) Scheduling Pods onto Nodes
- D) Providing container images

Correct Answer: B

Detailed Solution: etcd is a distributed key-value store used by Kubernetes to persist important cluster state, configuration, and metadata. The Kubernetes control plane relies on it as its backing data store.

QUESTION 5:

Which of the following is an example of a Type 2 (hosted) hypervisor?

- A) VMware ESXi
- B) Xen
- C) Oracle VirtualBox
- D) Microsoft Hyper-V Server

Correct Answer: C

Detailed Solution: Oracle VirtualBox is a Type 2 hypervisor because it runs as an application on top of an existing host operating system. VMware ESXi and Xen are examples of Type 1 hypervisors.

QUESTION 6:

A cloud data center has 100 physical servers, but only 40 are heavily utilized while the remaining servers operate at very low utilization. Which strategy would MOST effectively reduce energy consumption?

- A) Distribute VMs equally across all 100 servers
- B) Consolidate VMs onto fewer servers and switch idle servers to a low-power state
- C) Increase CPU frequency on all servers
- D) Duplicate every VM across multiple servers



Correct Answer: B

Detailed Solution: VM consolidation migrates workloads from lightly loaded servers onto fewer physical machines. Underutilized servers can then be switched to sleep or power-off states, reducing unnecessary energy consumption.

QUESTION 7:

In a Sensor Cloud environment, why is a middleware layer often required between Wireless Sensor Networks (WSNs) and cloud services?

- A) To increase the physical size of sensor nodes
- B) To abstract heterogeneous sensors and provide cloud-accessible sensor resources
- C) To eliminate all communication between sensors
- D) To replace the sensing function of WSN nodes

Correct Answer: B

Detailed Solution: A middleware layer provides an abstraction between resource-constrained, heterogeneous sensor networks and cloud platforms. It can handle resource discovery, data collection, protocol translation, virtualization/abstraction, and service management, allowing cloud applications to access sensor resources more conveniently.

QUESTION 8:

Which architecture is generally BEST suited for applications requiring both low latency and large-scale cloud analytics?

- A) IoT Device → Cloud only
- B) IoT Device → Edge/Fog → Cloud
- C) IoT Device → Database only
- D) IoT Device → Hypervisor → Cloud

Correct Answer: B

Detailed Solution: An Edge/Fog–Cloud architecture combines local processing for latency-sensitive tasks with cloud resources for large-scale storage, historical analysis, and computationally intensive workloads.



QUESTION 9:

An IoT application sends sensitive sensor data through public networks to a cloud platform. Which mechanism is MOST important for protecting the data during transmission?

- A) Data encryption
- B) Increasing CPU frequency
- C) Increasing sensor sampling rate
- D) VM consolidation

Correct Answer: A

Detailed Solution: Encryption protects data while it travels between IoT devices, gateways, and cloud services. Secure communication protocols such as TLS can help prevent unauthorized parties from reading or modifying data in transit.

QUESTION 10:

Statement 1: Processing latency-sensitive IoT workloads at an edge node can reduce the communication delay associated with a distant cloud.

Statement 2: Edge processing eliminates all network latency because no communication is required.

- A) Statement 1 is True and Statement 2 is False
- B) Statement 2 is True and Statement 1 is False
- C) Both statements are True
- D) Both statements are False

Correct Answer: A

Detailed Solution: Edge computing reduces latency by moving processing closer to the data source, but some communication latency still exists. Therefore, it does not eliminate network latency completely.